Towards Socially-Aware Modeling of Surgical Teams via Time-Expanded Graphs

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What occurs in Operating rooms (ORs) is typically analyzed from a procedural perspective, with a focus on surgery phases, technical execution, and workflow optimization. In contrast, the role played by social dynamics in shaping team performance remains overlooked. To address this gap, we propose Time-Expanded Graphs (TEG) in which temporal evolution collapses into a single graph. In TEG, nodes encode team members in a specific time step and typed edges encode temporal and relational dependencies among the team members.

We carried out experiments building on an existing dataset of audio-video recordings of simulated surgical procedures [1], and we enriched it with a set of automatically extracted features and manual annotations. Specifically, we integrated: (i) sixteen node-level behavioral descriptors capturing leadership and teamwork, (ii) five edge-level descriptors modeling interaction quality, and (iii) six graph-level annotations associated with teamwork performance and procedural duration. Moreover, questionnaires to identify counterfactuals were designed to support intervention analysis. Through our approach, we propose a unified learning setting that jointly captures team performance and counterfactual reasoning signals. Features values are aggregated over six minute consecutive windows of recordings. The same time span was used to manually annotate the excerpts. We address both efficiency prediction through TEG Neural Networks and interpretability through a two-layer counterfactual reasoning, represented in Fig. 1: (i) individual-level interventions on node attributes representing interpretable behavioral traits, (ii) interaction-level interventions modifying communication patterns. We explore two complementary strategies for counterfactual generation. The first is a greedy search approach, explicitly evaluated with respect to surgical duration [2] (i.e., improving slow vs. fast procedure predictions). The second is an explanation-driven approach [3], guided by graph-level information to identify influential structures and interactions.

Overall, our work enables both accurate prediction of team performance and prescriptive analysis through multi-level counterfactual reasoning, while consistently outperforming competitive graph-based, temporal, and tempo-relational baselines across efficiency prediction. This bridges graph-based machine learning with social and behavioral sciences in high-stakes collaborative environments.

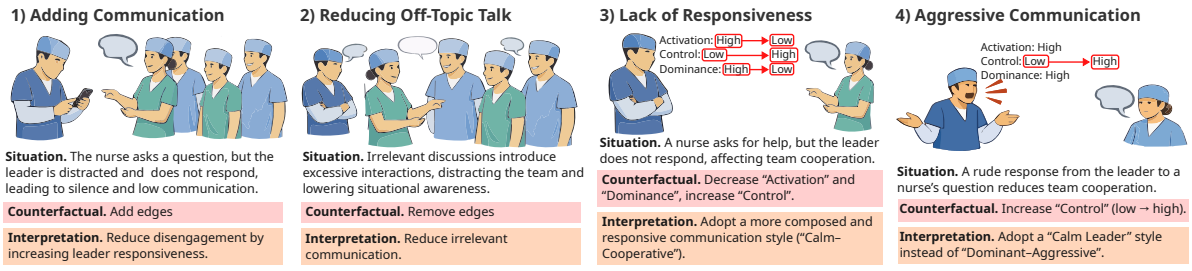


Figure 1: Counterfactual interventions on team communication dynamics. Modifying individual behaviors, interaction patterns, and temporal evolution enables improved prediction and efficiency of surgical procedures.

References

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